

Milestone 1

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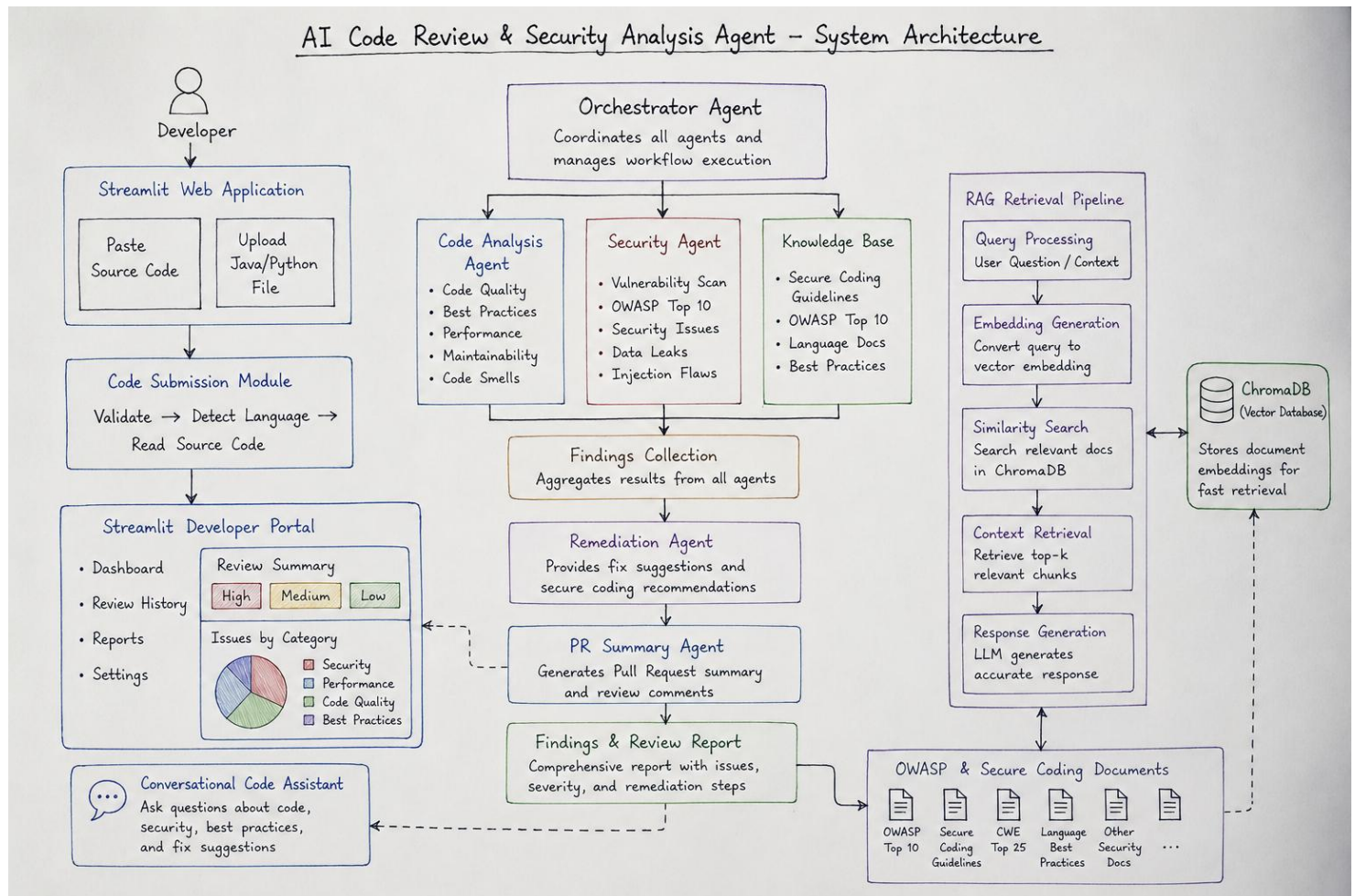
Git hub repo: https://github.com/Rajashekar25-colud/AI_Code_Review_And_Security_Analysis_Agent

Website: <https://aicodereviewandsecurityanalysisagent-nhndxuyepv8368zmzk39kx.streamlit.app/>

Project Title: AI Code Review & Security Analysis Agent

System Architecture

The AI Code Review & Security Analysis Agent follows a multi-agent architecture where each agent is responsible for a single specialized task. An Orchestrator Agent coordinates communication between all agents and ensures a smooth execution flow.



Agent Responsibilities

1. Code Analysis Agent

Purpose: Analyze source code quality and maintainability.

Responsibilities

- Detect Long Methods
- Detect Duplicate Code
- Detect Large Classes
- Detect Poor Variable Names
- Detect Magic Numbers
- Detect Dead Code
- Detect High Cyclomatic Complexity
- Detect Deep Nesting
- Generate Quality Score

2. Security Vulnerability Agent

Purpose: Identify OWASP-standard security vulnerabilities.

Responsibilities

- SQL Injection
- Cross-Site Scripting (XSS)
- Cross-Site Request Forgery (CSRF)
- Hardcoded Secrets
- Weak Authentication
- Broken Access Control
- Insecure File Handling
- Command Injection
- Sensitive Data Exposure

3. Remediation Agent

Purpose:Generate secure and optimized fixes.

Responsibilities

- Suggest corrected code
- Explain vulnerabilities
- Recommend secure coding practices
- Recommend design improvements
- Provide OWASP references

4. PR Summary Agent

Purpose: Create a developer-friendly review report.

Responsibilities

- Overall Summary
- Security Score
- Quality Score
- High Severity Count
- Medium Severity Count
- Low Severity Count
- Improvement Roadmap

5. Conversational Code Assistant

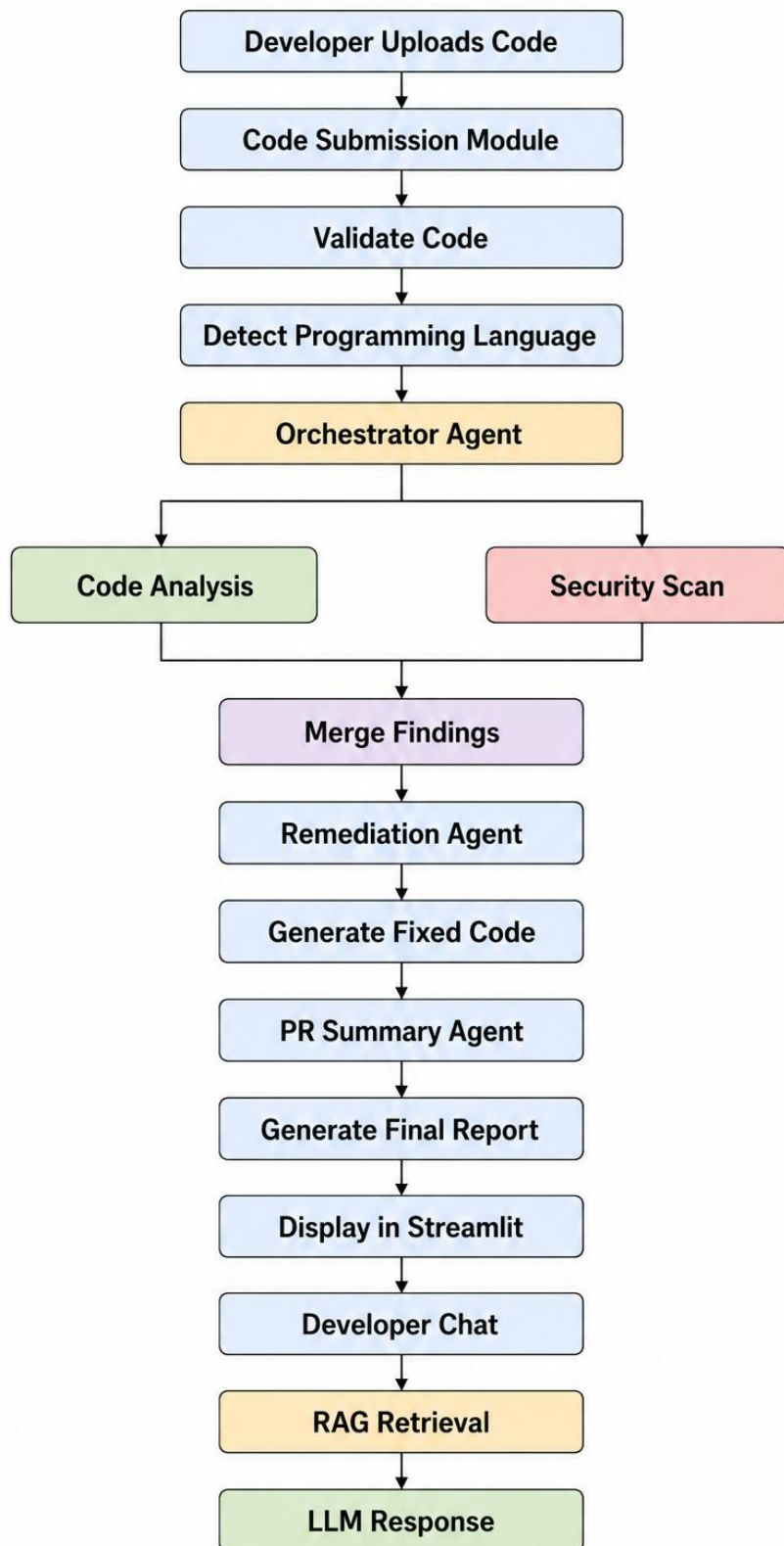
Purpose: Provide interactive developer guidance.

Responsibilities

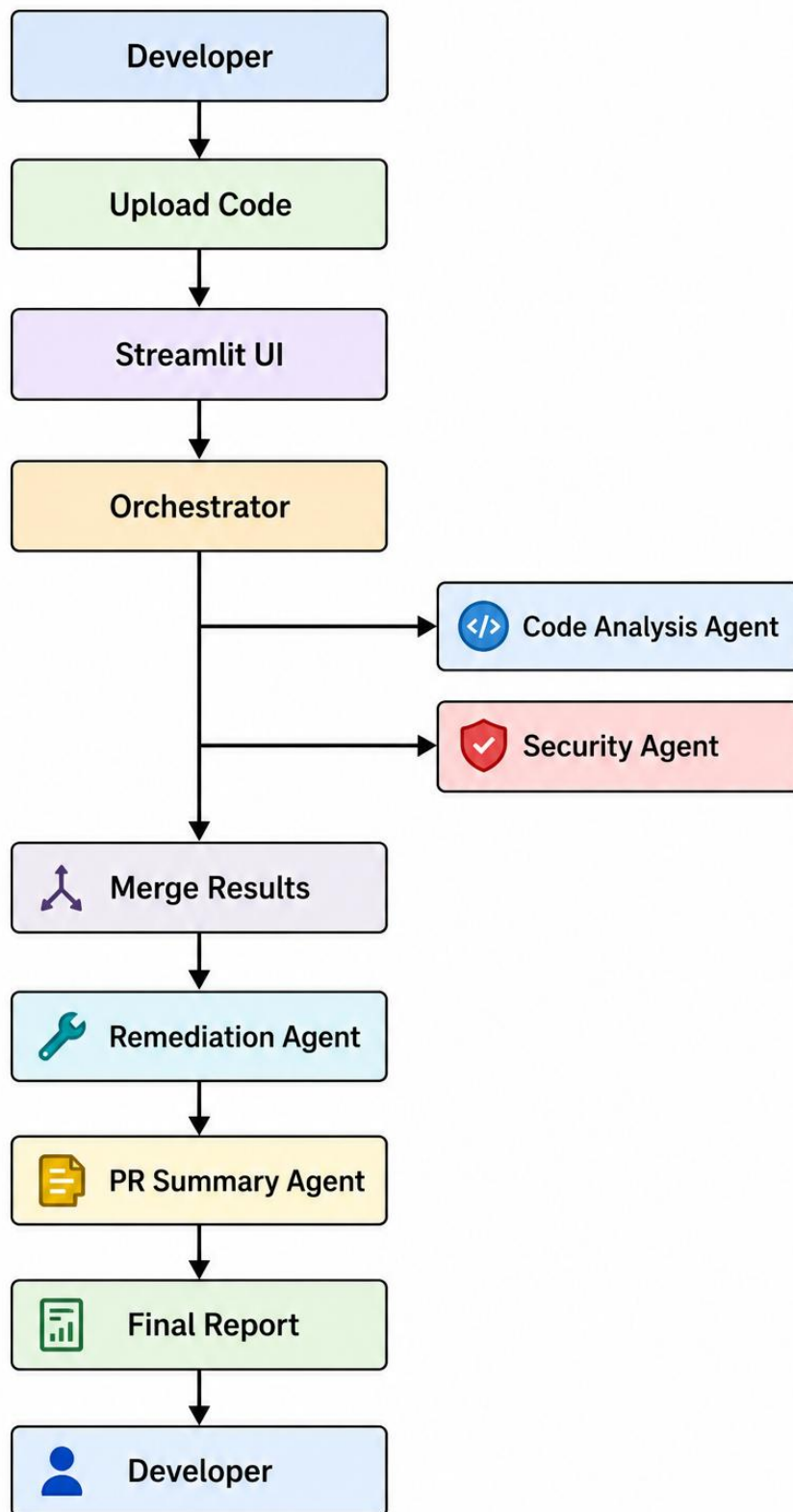
- Answer follow-up questions
- Explain vulnerabilities
- Explain secure coding
- Explain best practices
- Retrieve relevant OWASP information using RAG

Orchestration Flow

The Orchestrator Agent manages the complete execution pipeline.

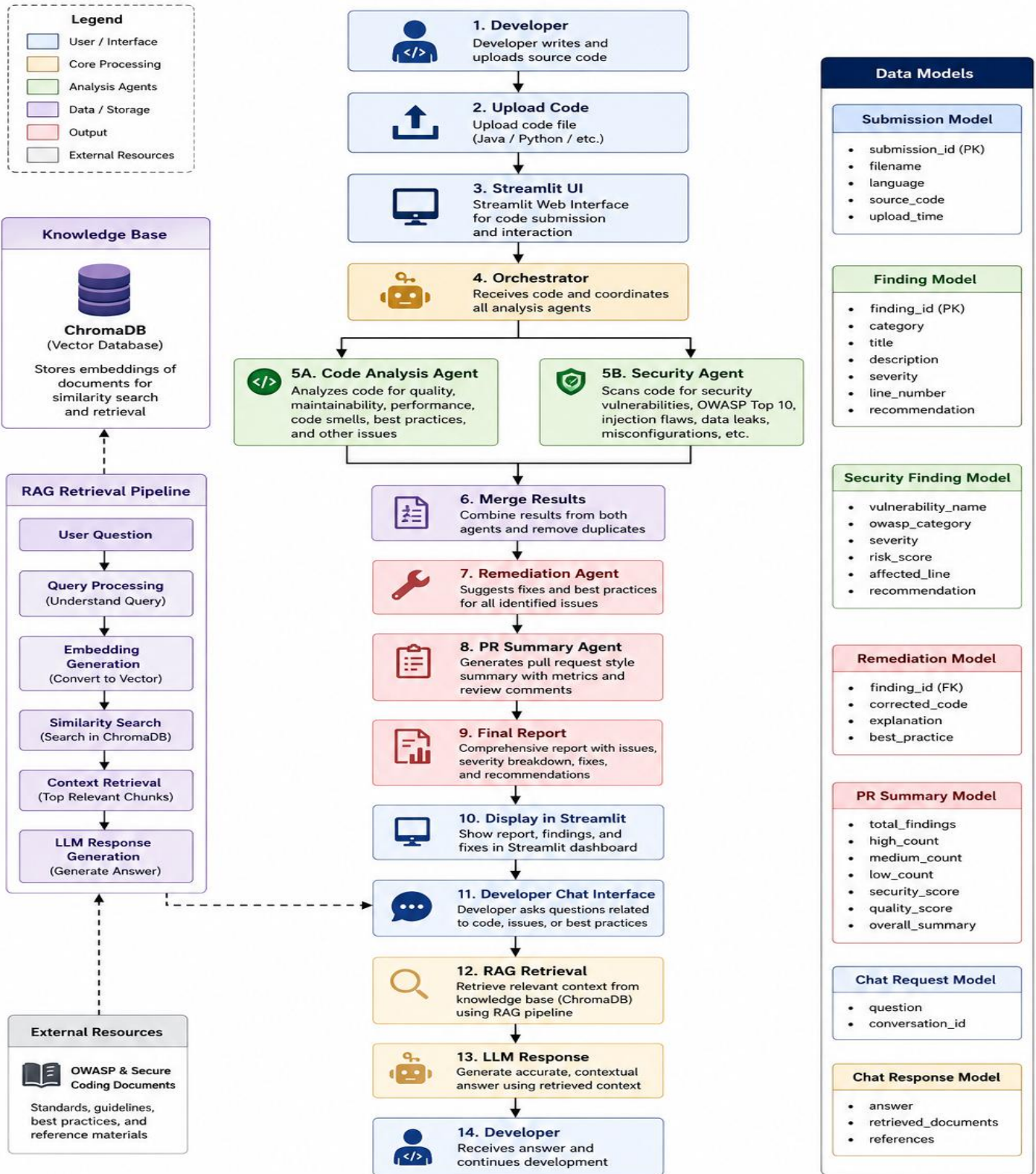


Sequence Flow

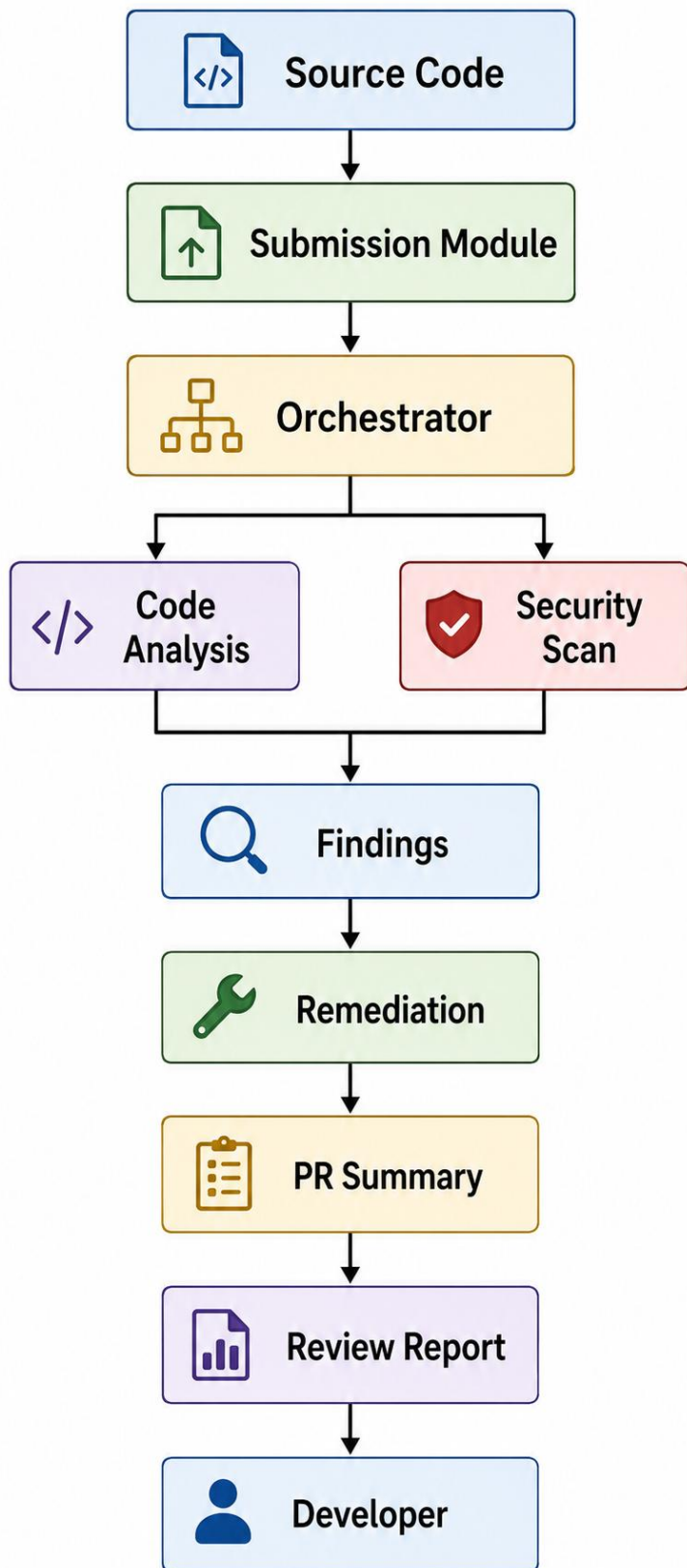


Data Models

AI Code Review & Security Analysis Agent – System Architecture



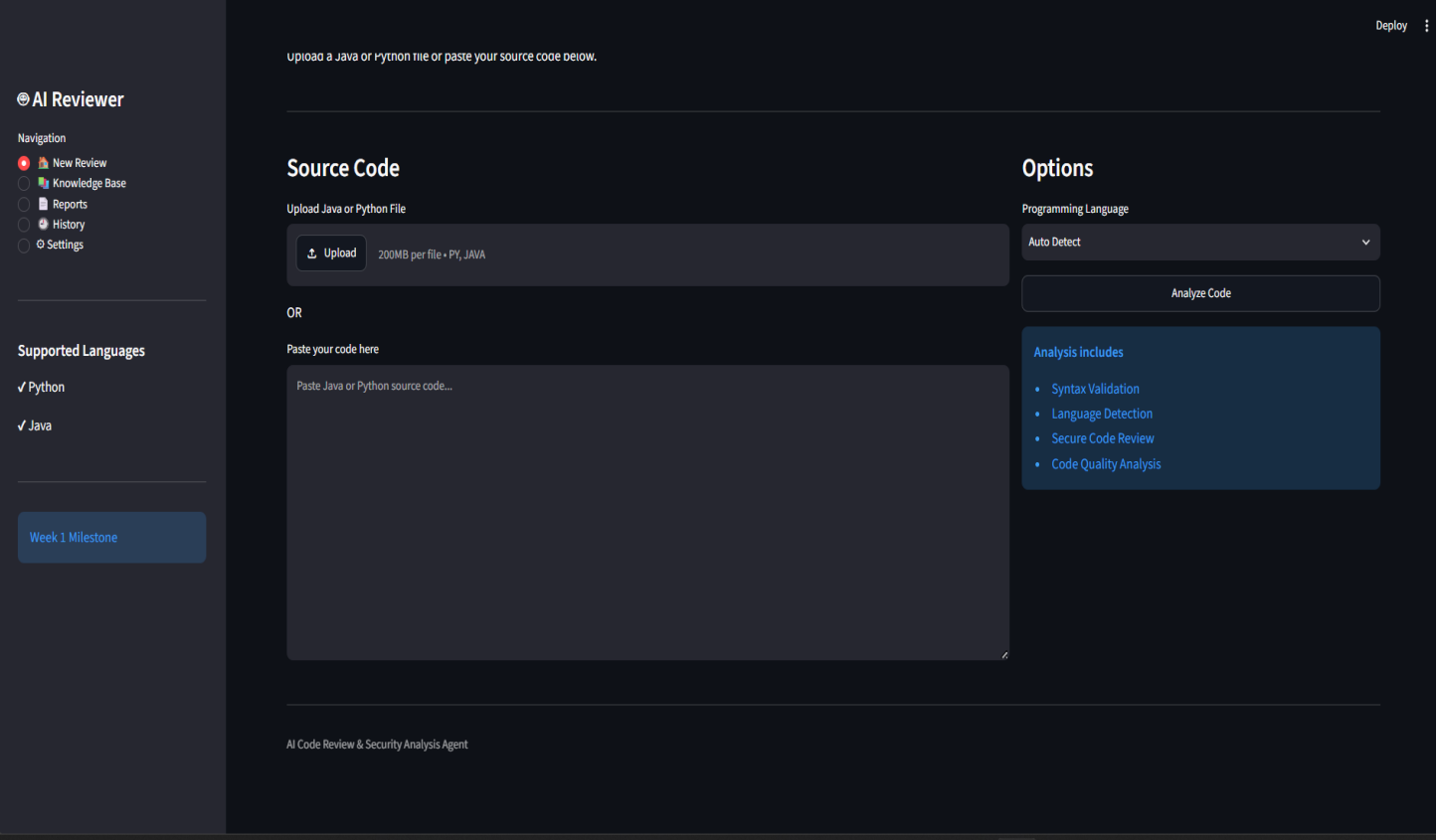
Data Flow Diagram



Design Advantages

- Modular multi-agent architecture improves maintainability.
- Independent agents allow parallel execution of quality and security analysis.
- RAG-based conversational assistant provides context-aware answers grounded in OWASP and secure coding documents.
- Streamlit offers a simple, interactive developer interface.
- ChromaDB enables fast semantic retrieval from the secure coding knowledge base.
- Structured data models ensure consistent communication between agents and simplify future extension of the platform.

OUT PUTS for task 3 and Task 4



AI Reviewer

Navigation

- New Review
- Knowledge Base
- Reports
- History
- Settings

Supported Languages

- Python
- Java

StackArray.java1.1KB

OR

Paste your code here

Paste Java or Python source code...

Auto Detect

Analyze Code

Analysis includes

- Syntax Validation
- Language Detection
- Secure Code Review
- Code Quality Analysis

Analysis Result

File Name : StackArray.java

Language : Java

Total Lines : 54

Java syntax is valid.

Status : Ready for Security Analysis

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- Java

hello.py19.0B

OR

Paste your code here

Paste Java or Python source code...

Auto Detect

Analyze Code

Analysis includes

- Syntax Validation
- Language Detection
- Secure Code Review
- Code Quality Analysis

Analysis Result

File Name : hello.py

Language : Python

Total Lines : 1

Python syntax is valid.

Status : Ready for Security Analysis

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Deploy

Secure Coding Knowledge Base

The following documents are indexed.

- OWASP Top 10
- Java Secure Coding
- Python Secure Coding
- Secure Coding Guidelines

Build Knowledge Base

Knowledge Base Created Successfully!

AI Code Review & Security Analysis Agent

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OR

Paste your code here

print("Hello")

Programming Language

Auto Detect

Analyze Code

Analysis includes

- Syntax Validation
- Language Detection
- Secure Code Review
- Code Quality Analysis

Analysis Result

File Name : Pasted Code

Language : Python

Total Lines : 1

Python syntax is valid.